

Lecture 5: Gauss-Markov Theorem

Economics 326 — Introduction to Econometrics II

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There are many alternative estimators

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We use the notation $E[\cdot | \mathbf{X}] = E[\cdot | X_1, \dots, X_n]$.

- The OLS estimator is not the only estimator we can construct. There are alternative estimators with some desirable properties.
- Example: Using only the first two observations, suppose that $X_2 \neq X_1$.

$$\tilde{\beta} = \frac{Y_2 - Y_1}{X_2 - X_1}.$$

- $\tilde{\beta}$ is **linear**:

$$\tilde{\beta} = c_1 Y_1 + c_2 Y_2,$$

where

$$c_1 = -\frac{1}{X_2 - X_1} \text{ and } c_2 = \frac{1}{X_2 - X_1}.$$

Unbiasedness of $\tilde{\beta}$

- If $Y_i = \alpha + \beta X_i + U_i$ and $E[U_i | \mathbf{X}] = 0$, then $\tilde{\beta}$ is **unbiased**:

$$\begin{aligned} \tilde{\beta} &= \frac{Y_2 - Y_1}{X_2 - X_1} \\ &= \frac{(\alpha + \beta X_2 + U_2) - (\alpha + \beta X_1 + U_1)}{X_2 - X_1} \\ &= \frac{\beta(X_2 - X_1)}{X_2 - X_1} + \frac{U_2 - U_1}{X_2 - X_1} \\ &= \beta + \frac{U_2 - U_1}{X_2 - X_1}, \text{ and} \end{aligned}$$

$$\begin{aligned} E[\tilde{\beta} | X_1, X_2] &= \beta + E\left[\frac{U_2 - U_1}{X_2 - X_1} | X_1, X_2\right] \\ &= \beta + \frac{E[U_2 | X_1, X_2] - E[U_1 | X_1, X_2]}{X_2 - X_1} \\ &= \beta. \end{aligned}$$

An optimality criterion

- Among all **linear** and **unbiased** estimators, an estimator with the **smallest variance** is called the **Best Linear Unbiased Estimator (BLUE)**.
- Note that the statement is **conditional** on **X**:
 - The estimators are **unbiased** conditionally on **X**.
 - The **variance** is conditional on **X**.

Gauss-Markov Theorem

Suppose that

- $Y_i = \alpha + \beta X_i + U_i$.
- $E[U_i | \mathbf{X}] = 0$.
- $E[U_i^2 | \mathbf{X}] = \sigma^2$ for all $i = 1, \dots, n$ (homoskedasticity).
- For all $i \neq j$, $E[U_i U_j | \mathbf{X}] = 0$.

Then, conditionally on **X**, the OLS estimators are **BLUE**.

Gauss-Markov Theorem (setup)

- We already know that the OLS estimator $\hat{\beta}$ is linear and unbiased (conditionally on **X**).
- Let $\tilde{\beta}$ be any other estimator of β such that
 - $\tilde{\beta}$ is linear:

$$\tilde{\beta} = \sum_{i=1}^n c_i Y_i,$$

where c 's depend only on **X**.

- $\tilde{\beta}$ is unbiased:

$$E[\tilde{\beta}] = \beta,$$

where expectation is conditional on **X**.

- We need to show that for **any** such $\tilde{\beta} \neq \hat{\beta}$,

$$\text{Var}(\tilde{\beta}) > \text{Var}(\hat{\beta}),$$

where the variance is conditional on **X**.

An outline of the proof

1. First, we are going to show that the c 's in $\tilde{\beta} = \sum_{i=1}^n c_i Y_i$ satisfy $\sum_{i=1}^n c_i = 0$ and $\sum_{i=1}^n c_i X_i = 1$.
2. Using the results of Step 1, we will show that conditionally on **X**, $\text{Cov}(\tilde{\beta}, \hat{\beta}) = \text{Var}(\hat{\beta})$.
3. Using the results of Step 2, we will show that conditionally on **X**, $\text{Var}(\tilde{\beta}) \geq \text{Var}(\hat{\beta})$.
4. Lastly, we will show that $\text{Var}(\tilde{\beta}) = \text{Var}(\hat{\beta})$ if and only if $\tilde{\beta} = \hat{\beta}$.

Proof: Step 1

- Since $\tilde{\beta} = \sum_{i=1}^n c_i Y_i$,

$$\begin{aligned}\tilde{\beta} &= \sum_{i=1}^n c_i (\alpha + \beta X_i + U_i) \\ &= \alpha \sum_{i=1}^n c_i + \beta \sum_{i=1}^n c_i X_i + \sum_{i=1}^n c_i U_i.\end{aligned}$$

- Conditionally on $\mathbf{X}$,

$$\begin{aligned}\mathbb{E} [\tilde{\beta}] &= \mathbb{E} \left[\alpha \sum_{i=1}^n c_i + \beta \sum_{i=1}^n c_i X_i + \sum_{i=1}^n c_i U_i \right] \\ &= \alpha \sum_{i=1}^n c_i + \beta \sum_{i=1}^n c_i X_i + \sum_{i=1}^n c_i \mathbb{E} [U_i] \\ &= \alpha \sum_{i=1}^n c_i + \beta \sum_{i=1}^n c_i X_i.\end{aligned}$$

Proof: Step 1 (continued)

- From the **linearity** we have that, conditionally on $\mathbf{X}$,

$$\mathbb{E} [\tilde{\beta}] = \alpha \sum_{i=1}^n c_i + \beta \sum_{i=1}^n c_i X_i.$$

- From the **unbiasedness** we have that conditionally on $\mathbf{X}$,

$$\beta = \mathbb{E} [\tilde{\beta}] = \alpha \sum_{i=1}^n c_i + \beta \sum_{i=1}^n c_i X_i.$$

- Since this has to be true for **any** α and β , it follows now that

$$\sum_{i=1}^n c_i = 0, \quad \sum_{i=1}^n c_i X_i = 1.$$

Proof: Step 2

- We have

$$\tilde{\beta} = \beta + \sum_{i=1}^n c_i U_i, \quad \text{with } \sum_{i=1}^n c_i = 0, \sum_{i=1}^n c_i X_i = 1.$$

$$\hat{\beta} = \beta + \sum_{i=1}^n w_i U_i, \quad \text{with } w_i = \frac{X_i - \bar{X}}{\sum_{j=1}^n (X_j - \bar{X})^2}.$$

- Conditionally on $\mathbf{X}$,

$$\begin{aligned}\text{Cov} (\tilde{\beta}, \hat{\beta}) &= \mathbb{E} [(\tilde{\beta} - \beta)(\hat{\beta} - \beta)] \\ &= \mathbb{E} \left[\left(\sum_{i=1}^n c_i U_i \right) \left(\sum_{i=1}^n w_i U_i \right) \right] \\ &= \sum_{i=1}^n c_i w_i \mathbb{E} [U_i^2] + \sum_{i=1}^n \sum_{j \neq i} c_i w_j \mathbb{E} [U_i U_j].\end{aligned}$$

Proof: Step 2 (continued)

$$\text{Cov}(\tilde{\beta}, \hat{\beta}) = \sum_{i=1}^n c_i w_i \text{E}[U_i^2] + \sum_{i=1}^n \sum_{j \neq i} c_i w_j \text{E}[U_i U_j].$$

- Since $\text{E}[U_i^2] = \sigma^2$ for all i 's:

$$\sum_{i=1}^n c_i w_i \text{E}[U_i^2] = \sigma^2 \sum_{i=1}^n c_i w_i.$$

- Since $\text{E}[U_i U_j] = 0$ for all $i \neq j$,

$$\sum_{i=1}^n \sum_{j \neq i} c_i w_j \text{E}[U_i U_j] = 0.$$

- Thus,

$$\text{Cov}(\tilde{\beta}, \hat{\beta}) = \sigma^2 \sum_{i=1}^n c_i w_i.$$

Proof: Step 2 (continued)

Conditionally on $\mathbf{X}$:

$$\text{Cov}(\tilde{\beta}, \hat{\beta}) = \sigma^2 \sum_{i=1}^n c_i w_i \text{ and } w_i = \frac{X_i - \bar{X}}{\sum_{j=1}^n (X_j - \bar{X})^2}.$$

$$\begin{aligned} \text{Cov}(\tilde{\beta}, \hat{\beta}) &= \sigma^2 \sum_{i=1}^n c_i \frac{X_i - \bar{X}}{\sum_{j=1}^n (X_j - \bar{X})^2} \\ &= \frac{\sigma^2}{\sum_{j=1}^n (X_j - \bar{X})^2} \sum_{i=1}^n c_i (X_i - \bar{X}) \\ &= \frac{\sigma^2}{\sum_{j=1}^n (X_j - \bar{X})^2} \left(\sum_{i=1}^n c_i X_i - \bar{X} \sum_{i=1}^n c_i \right) \\ &= \frac{\sigma^2}{\sum_{j=1}^n (X_j - \bar{X})^2} (1 + \bar{X} \cdot 0) \\ &= \text{Var}(\hat{\beta}). \end{aligned}$$

Proof: Step 3

- We know now that for any **linear** and **unbiased** $\tilde{\beta}$,

$$\text{Cov}(\tilde{\beta}, \hat{\beta}) = \text{Var}(\hat{\beta}).$$

- Let's consider $\text{Var}(\tilde{\beta} - \hat{\beta})$:

$$\begin{aligned} \text{Var}(\tilde{\beta} - \hat{\beta}) &= \text{Var}(\tilde{\beta}) + \text{Var}(\hat{\beta}) - 2\text{Cov}(\tilde{\beta}, \hat{\beta}) \\ &= \text{Var}(\tilde{\beta}) + \text{Var}(\hat{\beta}) - 2\text{Var}(\hat{\beta}) \\ &= \text{Var}(\tilde{\beta}) - \text{Var}(\hat{\beta}). \end{aligned}$$

- But since $\text{Var}(\tilde{\beta} - \hat{\beta}) \geq 0$,

$$\text{Var}(\tilde{\beta}) - \text{Var}(\hat{\beta}) \geq 0$$

or

$$\text{Var}(\tilde{\beta}) \geq \text{Var}(\hat{\beta}).$$

Proof: Step 4 (Uniqueness)

Suppose that $\text{Var}(\tilde{\beta}) = \text{Var}(\hat{\beta})$.

- Then,

$$\text{Var}(\tilde{\beta} - \hat{\beta}) = \text{Var}(\tilde{\beta}) - \text{Var}(\hat{\beta}) = 0.$$

- Thus, $\tilde{\beta} - \hat{\beta}$ is not random or

$$\tilde{\beta} - \hat{\beta} = \text{constant}.$$

- This constant also has to be zero because

$$\begin{aligned} \text{E}[\tilde{\beta}] &= \text{E}[\hat{\beta}] + \text{constant} \\ &= \beta + \text{constant}, \end{aligned}$$

and in order for $\tilde{\beta}$ to be **unbiased**

$$\text{constant} = 0 \text{ or } \tilde{\beta} = \hat{\beta}.$$